Roll No.: _____

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Missed Mid-Term Examinations – October 2025

Third Semester

Artificial Intelligence & Data Science

23MAT204 MATHEMATICS FOR INTELLIGENT SYSTEMS III

Duration: Two hours

Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Apply the fundamental techniques of optimization theory to solve data science problems.
CO02	Analyse and solve computationally, physical systems using the formalism of partial differential equations.
CO03	Apply Markovian concepts in stochastic sequential systems.
CO04	Explain Bells Inequality and Quantum gates.

Answer all questions

Group A: Descriptive Type Questions

1. Use the method of Lagrange multipliers to find the dimensions of an open rectangular box (top open) of maximum volume that can be constructed from a sheet of metal with an area of 48square units. [5][CO01][BTL4]

2. A furniture manufacturer makes two products: Chairs (C)and Tables (T). Each Chair requires 2hours of carpentry and 1hour of painting. Each Table requires 3hours of carpentry and 2hours of painting. The manufacturer has at most 24hours of carpentry time and 16hours of painting time available per day. The profit is Rs. 400per Chair and Rs. 700per Table. Tasks: Formulate this situation as a Linear Programming (LP) problem — define decision variables, write the objective function, and list the constraints. [3][CO01][BTL6]

3. Solve the LP problem formulated in Question 2 to find the optimal production quantities of Chairs and Tables that maximize daily profit, and compute that maximum profit. (Hint: Use the graphical method.) [2][CO01][BTL 3]

4. Linearity and Convexity: (a) Check convexity of the set $S = \{(x, y): y \geq x^2\}$. Determine whether $S$is convex and justify your answer. (b) Check linearity of the functions $f_1(x, y) = 4x - 7y$and $f_2(x, y) = x^2 + y$. Determine whether each is linear; justify using additivity and homogeneity. [2+3][CO01][BTL2]

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5. (i) Find the differential equation of all spheres of fixed radius $r$whose centers lie on the z-axis. (ii) Eliminate the arbitrary function $f$from $z = e^{ny} f(x - y)$to obtain the partial differential equation. [3+2][CO02][BTL3]

6. (i) Distinguish between linear, semi-linear, and quasi-linear partial differential equations, giving one example of each. (ii) Solve the PDE $z = px + qy + p^2 + q^2$(Clairaut's form) to find the complete and general solution. [3+2][CO02][BTL 1,5]

7. Classify each of the following partial differential equations as elliptic, parabolic, or hyperbolic: (a) $3u_{xx} + u_{yy} = 0$; (b) $u_{xx} - 2u_{xy} + u_{yy} = 0$; (c) $u_{tt} - 16u_{xx} = 0$. [5][CO02][BTL 2]

8. Let Sbe the sample space of tossing three fair coins simultaneously. Define random variables X: the number of heads obtained; Y: the absolute difference between the number of heads and tails, $Y = |H - T|$. Find the probability distributions of $X$and Y. Also compute their expectations $E[X]$, $E[Y]$and variances $\mathrm{Var}(X)$, $\mathrm{Var}(Y)$. [10][CO03][BTL3]

Group B: Multiple Choice Questions

9. The PDE $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$represents:
(A) One-dimensional Heat Equation; (B) One-dimensional Wave Equation; (C) Two-dimensional Laplace Equation; (D) Poisson Equation. [1][CO02][BTL1]

10. Which condition ensures that the PDE $Au_{xx} + Bu_{xy} + Cu_{yy} + \cdots = 0$is hyperbolic?
(A) $B^2 - 4AC < 0$; (B) $B^2 - 4AC = 0$; (C) $B^2 - 4AC > 0$; (D) $A = C$. [1][CO02][BTL1]

11. A string of length $L$is fixed at both ends ($x = 0$ and $x = L$). If the string is released from rest with an initial displacement $f(x)$, the boundary conditions are:
(A) $u(0,t) = 0$, $u(L,t) = 0$; (B) $u_x(0,t) = 0$, $u_x(L,t) = 0$;
(C) $u(x,0) = 0$, $u_t(x,0) = 0$; (D) $u(0,t) = L$, $u(L,t) = 0$. [1][CO02][BTL 2]

12. If $z = \ln(x^2 + y^2)$, then $z_{xx} + z_{yy}$equals:
(A) 1; (B) 0; (C) $x + y$; (D) $x^2 + y^2$. [1][CO02][BTL2]

13. In the finite difference method, the forward difference approximation of the first derivative dy/dxat point $x_i$is: (A) $\frac{y_{i+1} - y_i}{h}$; (B) $\frac{y_i - y_{i-1}}{h}$; (C) $\frac{y_{i+1} - y_{i-1}}{2h}$; (D) $\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2}$. [1][CO02][BTL 1]

14. Independent random variables $X$and Yhave variances 3and 4respectively. Let $Z = 2X - 3Y$. The variance of $Z$is: (A) -6; (B) 6; (C) 48; (D) 24. [1][CO03][BTL 1]

15. The expected value $E[X]$of a random variable $X$with distribution below is:

X	**-1**	**0**	**1**	**2**
P(X)	0.2	0.3	0.1	0.4

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Options: (A) 0.5; (B) 0.7; (C) 0.9; (D) 1.0. [1][CO03][BTL 3]

16. Two random variables $X$and $Y$are said to be independent if:
(A) $E[XY] = 0$; (B) $E[XY] = E[X] + E[Y]$; (C) $P(X \cap Y) = P(X) \cdot P(Y)$; (D) $\text{Cov}(X, Y) = 1$.

[1][CO03][BTL4

17. Which function is convex but non-linear?
(A) $f(x) = 2x + 5$; (B) $f(x) = x^2$; (C) $f(x) = \sin x$; (D) $f(x) = -|x|$.

[1][CO01][BTL2]

18. In a Linear Programming Problem, the set of all feasible solutions (the feasible region) forms a:
(A) Concave set; (B) Convex set; (C) Null set; (D) Unbounded set always. [1][CO01][BTL 1]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**17**	BTL 1	**8**
CO02	**20**	BTL 2	**13**
CO03	**13**	BTL 3	**18**
		BTL 4	**6**
		BTL 5	**2**
		BTL 6	**3**

Bloom's Taxonomy Levels (attached for reference)

Level 1 – Remember

Level 2 - Understand

Level 3 – Apply

Level 4 - Analyze

Level 5 - Evaluate

Level 6- Create

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